Karan

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SUMMARY

Final-year B.Tech CSE student and full-stack engineer who builds AI-powered products end to end in TypeScript, React, Node.js and PostgreSQL. Built multi-agent LLM workflows, rubric-based grading, document pipelines with Qdrant vector search and tool-calling agents (text and Realtime voice). Shipped an AI exam platform used by 20+ real students and an AWS-hackathon app covered by 15 automated API tests.

WORK EXPERIENCE

Coding Instructor, Hartron Skill Center, Pehowa Jan 2025 – Present

- Taught programming fundamentals to 100+ students across C, C++ and web development basics, with structured, well-documented lesson plans and hands-on exercises.
- Ran regular doubt-clearing sessions, diagnosing learners' problems and explaining fixes clearly, and used assessments to track progress and improve concept retention across batch cycles.

PROJECTS

Abhyas – AI-Powered Examination and Assignment Platform

Tech Stack: Node.js, Express.js, TypeScript, PostgreSQL (Neon), Drizzle ORM, Inngest, BullMQ, Redis, OpenAI API, OpenAI Realtime, Qdrant, Python (PyMuPDF), Next.js, React, Cloudinary, Render, Vercel

- Built the backend of a multi-tenant AI exam platform with role-based access (students, teachers, organisation managers, system admins), scheduled and on-demand exams delivered via join codes, assignments, and subscription plans with per-organisation usage quotas. Used by 20+ real students.
- Designed a multi-agent exam-generation pipeline: a conversational intent agent collects the teacher's preferences, a subtopic agent plans coverage and weights, a deterministic allocator distributes question counts, and a generation agent writes questions per topic with Zod-validated structured outputs. Blueprint and question generation run as retryable Inngest jobs, with topics generated in parallel.
- Built tool-calling review agents (text and OpenAI Realtime voice) that edit the exam blueprint and live question list through schema-validated tools, with deterministic code applying every change and independent fact-checking of each MCQ's marked answer before it is saved.
- Built rubric-based AI grading for descriptive answers with category-wise scoring and personalized feedback, run through a BullMQ/Redis queue, behind a fail-closed guardrail agent that screens submissions for prompt injection.
- Built a PYQ ingestion pipeline (Inngest, PyMuPDF) that extracts and classifies questions from PDFs, including MCQ options, answer keys, tables and images; generates embeddings; stores them in Qdrant; and serves semantic search scoped to the owner or organisation.
- Designed a vector-less RAG pipeline for textbooks: an LLM builds a hierarchical chapter/section index from 10-page windows, and sections are retrieved by citation-based matching to ground generated questions in real book content.

SmartBin – AI-Verified Dustbin Finder (Team Lead, AWS Hackathon)

Tech Stack: TypeScript, Next.js, Node.js, Express.js, PostgreSQL, Drizzle ORM, OpenAI SDK, AWS (EC2, RDS, S3, Cognito, API Gateway)

- Led a team of 3 to build an app that shows the nearest public dustbins (wet or dry) with walking directions, where anyone can add a bin with a live-camera photo. Built the backend and the AWS deployment.
- Built the core of the app as an LLM call: an AI photo check (OpenAI vision, structured outputs) that accepts only real public dustbins. Measured on 20 labelled photos, it rejected all 12 look-alikes (buckets, post boxes, garbage piles) and reached 95% accuracy overall.
- Blocked fake entries with live-camera-only photos, real-image checks, a 15 m duplicate rule, monthly limits and location-checked missing reports, covered by 15 automated API tests.
- Deployed on AWS (Cognito, API Gateway, EC2, RDS); photos upload straight to a private S3 bucket and secrets stay encrypted in Systems Manager, with no access keys on the server.

Alhnkar – Full-Stack E-Commerce Platform

Tech Stack: React.js, Tailwind CSS, Zustand, Node.js, Express.js, MongoDB, Razorpay, Cloudinary, Resend

- Architected and deployed a full-stack e-commerce app with Razorpay checkout, including server-side signature verification to prevent fraudulent orders.
- Delivered a fully responsive, animation-rich storefront in React and Tailwind CSS with centralized global state in Zustand.
- Built backend modules for carts, orders, product reviews and shipment tracking, with JWT and bcrypt-based authentication.
- Integrated third-party APIs for payments (Razorpay), image storage (Cloudinary) and automated transactional emails (Resend).

SKILLS

Languages & Frontend: TypeScript, JavaScript, React.js, Next.js, Tailwind CSS, Zustand

Backend: Node.js, Express.js, REST API design, Zod validation, JWT and RBAC, Inngest and BullMQ/Redis (retryable background jobs and queues), third-party integrations (Razorpay, Cloudinary, Resend)

AI & LLM: Prompt engineering, agentic workflows, structured outputs, rubric-based evaluation, testing on labelled sets, prompt-injection guardrails, RAG, embeddings, Qdrant (vector DB), OpenAI API, OpenAI Realtime

Databases: PostgreSQL (schema design, migrations), Drizzle ORM, MongoDB, Mongoose

Cloud & Security: AWS (EC2, RDS, S3, Cognito, API Gateway, Systems Manager), Render, Vercel, Neon, HTTPS, bcrypt, AES-256 encryption

Testing & Tools: Automated API testing, Git, GitHub

EDUCATION

B.Tech in Computer Science and Engineering, SKIET College, Kurukshetra Aug 2023 – Jun 2027